

lambda_analysis.cxx

Code Reference Manual

EIC $\Lambda \rightarrow p + \pi^-$ Analysis — Far-Forward Region

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1 Overview

`lambda_analysis.cxx` is a ROOT macro that analyses $\Lambda \rightarrow p + \pi^-$ decays in the EIC far-forward region. It reads Monte Carlo truth from PODIO/EDM4hep files produced by the EIC simulation framework (dd4hep/GEANT4), fills histograms, and produces three output PDF files plus one ROOT file containing all histograms.

Initial version: Dmitry Romanov from the EIC Meson Structure Working Group https://github.com/JeffersonLab/meson-structure/blob/main/tutorials/cpp_01_read_edm4eic.cxx

1.1 How to run

Before running this macro, please read the instruction of how to install and run the EIC shell in <https://jeffersonlab.github.io/meson-structure/>.

THIS MACRO ONLY RUNS IN THE EIC SHELL

```
# Interactive ROOT session
root -x -l lambda_analysis.cxx
root [0] lambda_analysis()           // all events, all files
root [0] lambda_analysis(500)        // 500 events per file (quick test)

# Batch mode (no GUI, faster, recommended for full runs)
root -x -l -b -q lambda_analysis.cxx
```

1.2 Dependencies

Library	Purpose
PODIO (podioDict, podioRootIO)	Reading EDM4hep event frames
EDM4hep (libedm4hepDict)	MC particle data model
fmt	Formatted printing to stdout
ROOT (standard)	Histograms, canvases, files, TProfile

1.3 Output files

File	Contents
<code>lambda_analysis_18x275.root</code>	All histograms (reopen to replot)
<code>lambda_analysis_18x275.pdf</code>	Diagnostic + particle analysis plots
<code>tracker_study_18x275.pdf</code>	Tracker resolution study (12 pages)

2 Configuration Namespaces

All tunable parameters live in two namespaces at the top of the file. **No other part of the code needs to be changed to adjust cuts or tracker geometry.**

2.1 namespace Cuts

Constant	Value	Meaning
LAMBDA_PDG	3122	PDG code of the Λ baryon
LAMBDA_PRIMARY_IDX	4	MC particle index of primary Λ
LAMBDA_N_DAUGHTERS	2	Selects charged decay ($p + \pi^-$)
DECAY_Z_MIN	20800 mm	Last beamline magnet exit
DECAY_Z_MAX	35000 mm	ZDC face
ZDC_Z	35000 mm	ZDC face (used for projection)
CROSSING_ANGLE	25 mrad	Hadron beam crossing angle
PDG_PROTON	2212	PDG code of proton
PDG_PION_MINUS	-211	PDG code of π^-
PDG_NEUTRON	2112	PDG code of neutron
PDG_PION_ZERO	111	PDG code of π^0
N_FILES	20	Number of input ROOT files
FILE_PATTERN	path template	<code>fmt::format</code> pattern
OUTPUT_PDF	string	Base name for analysis PDF

`LAMBDA_PRIMARY_IDX = 4` is generator-dependent. If the simulation configuration changes, this index must be updated. Verify by running with a small sample and checking the `[NON-PRIMARY]` diagnostic output (see Section 5.2).

2.2 namespace Tracker

Constant	Value	Meaning
N_PLANES	4	Number of tracking planes
LENGTH	2000 mm	Total tracker length (plane 1 to plane 4)
STANDOFF	500 mm	Gap from ZDC face to last plane
SIGMA_X[]	0.05–1.0 mm	Five pitch scenarios
N_SC	5	Number of scenarios
LABELS []	strings	ROOT title labels per scenario
EFF_THRESH []	500, 1000, 2000 mm	Efficiency thresholds
N_THRESH	3	Number of thresholds
PRIMARY_THRESH	1	Index of primary threshold (1000 mm)
OUTPUT_PDF	string	Base name for tracker study PDF

3 Global State

`map<string, TH1*> H` — Global histogram registry. All histograms are stored here by name. Access via the typed helpers `h1()` and `h2()` (see Section 6).

`string current_file` — Set to the filename currently being processed by `process_file()`. Used in diagnostic print statements to identify which file contains anomalous events.

`Counters counters` — Global instance of the `Counters` struct. Incremented throughout `process_event()` and printed at the end of the run.

4 Global Functions (Physics Utilities)

4.1 `sigma_theta_rad(sigma_x_mm)`

Returns angular resolution [rad] assuming all $N = 4$ planes resolve the tracks (best case). Retained for documentation and the summary table. In the event loop, use `sigma_theta_eff()` instead.

4.2 `plane_z(ip)`

Returns the absolute z position [mm] of tracker plane `ip` (1-indexed). Planes are equally spaced:

$$z_1 = 32,500 \text{ mm}, \quad z_4 = 34,500 \text{ mm}$$

4.3 `sigma_theta_eff(sigma_x_mm, first_plane)`

Returns the effective angular resolution [rad] using only planes from `first_plane` to `N_PLANES`. Returns 10^9 (sentinel = unresolvable) when fewer than 2 planes are available.

<code>first_plane</code>	N_{eff}	L_{eff} (mm)	Relative σ_θ
1	4	2000	1.0 (best)
2	3	1333	1.15
3	2	667	1.58
4	1	0	∞

4.4 `PolarAngle(px, py, pz)`

Returns $\theta = \arccos(p_z/|\vec{p}|)$ in degrees.

4.5 `PhiAngle(px, py)`

Returns $\phi = \text{atan2}(p_y, p_x)$ in degrees.

4.6 `OpeningAngle(a, b)`

Returns the opening angle between two `DaughterKin` objects in degrees, from the dot product of unit momentum vectors.

5 Data Structures

5.1 `struct DaughterKin`

Holds kinematic quantities for one decay daughter projected onto the ZDC plane. Computed by `compute()`.

Member	Unit	Meaning
<code>theta</code>	degrees	Polar angle w.r.t. z -axis
<code>phi</code>	degrees	Azimuthal angle
<code>xAtZDC, yAtZDC</code>	mm	Position at ZDC face (lab frame)
<code>radius</code>	mm	$\sqrt{x^2 + y^2}$ in lab frame
<code>xBC, yBC</code>	mm	Position relative to hadron beam axis
<code>radiusBC</code>	mm	Distance from beam axis

The vertex offset (crossing angle) is applied inside `compute()` using the true decay vertex position (x_v, y_v, z_v) . `isSet()` returns `true` if `theta` $\neq 0$.

5.2 struct Counters

Tracks particle counts at each selection stage. Printed to stdout by `counters.print()` at the end of the run.

Member	Counts
<code>total_events</code>	Events read from all files
<code>lambda_escaped</code>	Primary Λ with no daughters
<code>lambda_before_magnet</code>	Primary Λ decaying before 20.8 m
<code>lambda_beyond_zdc</code>	Primary Λ decaying past 35 m
<code>lambda_field_free</code>	Primary Λ in field-free window (any channel)
<code>lambda_charged</code>	Field-free, charged channel $p + \pi^-$ [used]
<code>lambda_neutral</code>	Field-free, neutral channel $n + \pi^0$ [not used]
<code>lambda_other_ndau</code>	Field-free, material interactions ($N_{\text{dau}} > 2$)
<code>proton</code>	Protons found in charged channel
<code>pion_minus</code>	π^- found in charged channel
<code>neutron</code>	Neutrons (neutral channel, informational)
<code>pion_zero</code>	π^0 (neutral channel, informational)

6 Histogram Registry

6.1 Design

All histograms live in `map<string, TH1*> H`. Two typed accessors avoid casting at every fill site:

```
inline TH1F* h1(const string& name) { return static_cast<TH1F*>(H.at(
    name)); }
inline TH2F* h2(const string& name) { return static_cast<TH2F*>(H.at(
    name)); }
```

For TProfile histograms, cast explicitly at the fill site:

```
static_cast<TProfile*>(H.at("ResZ_rms_vsAlpha_sc0"))->Fill(alpha, dz);
```

6.2 How to add a new histogram

1. In `book_histograms()`: add one line:

```
H["MyHist"] = new TH1F("MyHist", "Title;X;Y", 100, 0, 100);
```

2. In `process_event()`: fill it where the quantity is computed:

```
h1("MyHist")->Fill(my_value);
```

3. In `plot()` or `plot_tracker_study()`: draw it.
4. It is automatically written to the ROOT file by `save_histograms()` — no extra step needed.

6.3 Complete histogram table

Lambda kinematics

Key	Type	Contents
LambdaEnd	TH1F	Decay endpoint z , all Λ with 2 parents [0–35 m]
LambdaEnd_1	TH1F	Decay endpoint [0–6 m]
LambdaEnd_2	TH1F	Decay endpoint [6–20.8 m]
LambdaEnd_3	TH1F	Decay endpoint, primary Λ , field-free, any channel
LambdaEnd_analysis	TH1F	Decay endpoint, primary, field-free, charged channel only. Integral = <code>counters.lambda_charged</code>
LambdaGenZ	TH1F	Production vertex z
LambdaZnoDaughters	TH1F	Endpoint of escaped Λ (no daughters)
LambdaAngleGen	TH1F	Polar angle at production

Proton daughters

Vertex_p, Endpoint_p, Theta_p, XYatZDC_p (TH2F, lab), XYatZDC_p_BC (TH2F, beam-centred), ZvsTheta_p, ZvsRadius_p.

Pion daughters

Same set with `_pi` suffix.

Neutral channel (informational)

Vertex_n, Endpoint_n, Theta_n, XYatZDC_n for neutrons; same with `_pi0` for π^0 .

Correlations

ThetaCorr (TH2F), RadiusSumVsZ (TH2F), DeltaR, OpenAngle, DeltaRvsZ (TH2F).

Tracker study: ΔR at planes (truth)

Key	Type	Contents
DeltaR_pl{1-4}	TH1F	Track separation at plane i , full range [0–250 mm]
DeltaR_pl{1-4}_zoom	TH1F	Track separation at plane i , zoomed [0–3 mm], 0.05 mm/bin. Shows sub-mm region where pitch matters

Tracker study: analytical (suffix _sc0 to _sc4)

Key	Type	Contents
SigmaZ_sc{i}	TH1F	σ_z distribution [0–5000 mm]
SigmaAlpha_sc{i}	TH1F	σ_α distribution [0–5 mrad]
SigmaZ_vsAlpha_sc{i}	TH2F	σ_z vs true α [0–15 mrad]
SigmaZ_vsD_sc{i}	TH2F	σ_z vs D_{eff}

Tracker study: smearing (suffix _sc0 to _sc4)

Key	Type	Contents
ResZ_sc{i}	TH1F	$z_{\text{reco}} - z_{\text{true}}$ residual
ResAlpha_sc{i}	TH1F	$\alpha_{\text{reco}} - \alpha_{\text{true}}$ residual
ResZ_vsAlpha_sc{i}	TH2F	Δz vs true α [0–15 mrad]. Use <code>ProjectionY(name,b,b)</code> to get the Δz distribution per α bin
ZReco_sc{i}	TH1F	Reconstructed vertex z [15–40 m]
SmearEff_thr{j}_sc{i}	TH1F	Pass/fail histogram for $ \Delta z < \text{EFF_THRESH}[j]$. <code>GetBinContent(2)/GetEntries()</code> = precision efficiency
ResZ_rms_vsAlpha_sc{i}	TProfile	TProfile(“S”) of Δz vs α . Filled but not used for plotting; plotting uses <code>ProjectionY</code> of <code>ResZ_vsAlpha</code> instead

7 Functions Reference

7.1 `book_histograms()`

Called once at startup. Creates every histogram in `H`. All booking is here — change ranges or binning only in this function.

7.2 `process_event(event, event_number)`

Main event processing function. Called once per MC event.

Execution flow:

1. Retrieve `MCParticles` from the PODIO frame.
2. Loop over all particles. For each Λ with 2 parents: fill `LambdaEnd` histograms and `LambdaAngleGen`. If the particle is in the field-free window and is not the primary Lambda (index $\neq 4$), print a diagnostic.
3. For all particles: fill `LambdaGenZ`.
4. Select primary Lambda (index = `LAMBDA_PRIMARY_IDX`). If no daughters: fill `LambdaZnoDaughters`, increment `lambda_escaped`, skip.
5. Apply z window cut. Increment `lambda_before_magnet` or `lambda_beyond_zdc` accordingly.
6. Increment `lambda_field_free`.
7. Check daughter PDGs to identify channel. Increment `lambda_neutral` + fill neutral histograms + skip, or `lambda_other_ndau` + skip, as appropriate.
8. For the charged channel: increment `lambda_charged`, fill `LambdaEnd_analysis`.
9. Daughter loop: compute `DaughterKin` for p and π^- , fill all daughter histograms.
10. If both daughters found: fill correlation histograms (`DeltaR`, `OpenAngle`, etc.).
11. Tracker study (if $\alpha > 10^{-6}$ rad):
 - (a) Fill `DeltaR_pl{i}` and `DeltaR_pl{i}_zoom` for each plane.
 - (b) For each scenario: find `first_plane` where $\Delta R > \sigma_x$. If none: skip.
 - (c) Compute `sigma_theta_eff`, D_{eff} , σ_z . Fill analytical histograms.
 - (d) Smear slopes with $\mathcal{G}(0, \sigma_\theta)$. Reconstruct z_{reco} via PCA. Set `reco_ok` flag only when the PCA succeeded (not the fallback).
 - (e) Fill `ResZ`, `ResAlpha`, `ResZ_vsAlpha`, `ZReco`. If `reco_ok`: also fill `SmearEff_thr{j}` and `ResZ_rms_vsAlpha`.

7.3 process_file(filename, events_limit)

Sets `current_file = filename`, opens one PODIO ROOT file, calls `process_event()` for each entry up to `events_limit` ($-1 = \text{all}$). Wrapped in try/catch.

7.4 plot()

Creates diagnostic and particle analysis canvases. Returns a `vector<TCanvas*>` for `save_plots()`.

7.5 save_plots(pages, output_name)

Writes a canvas vector to a multi-page PDF using ROOT's `Print("name.pdf") / Print("name.pdf") / Print("name.pdf")` pattern.

7.6 plot_tracker_study()

Creates all 12 tracker study canvases. Returns a `vector<TCanvas*>` for `save_tracker_study()`.

The 12 pages in order:

1. ΔR at each tracker plane (2×2)
2. ΔR zoomed overlay [0–3 mm], all planes
3. σ_z distributions overlay (analytical)
4. σ_α distributions overlay (analytical)
5. σ_z vs α per scenario (3×2)
6. σ_z vs D_{eff} per scenario (3×2)
7. Summary: mean σ_z + analytical efficiency curves
8. z_{reco} distributions + precision smearing efficiency
9. Δz residuals overlay
10. $\Delta\alpha$ residuals overlay
11. Δz vs α per scenario (3×2)
12. Validation: $\text{RMS}(\Delta z)$ vs analytical σ_z per α bin

7.7 save_tracker_study(pages)

Writes the tracker canvas vector to `tracker_study_18x275.pdf`.

7.8 save_histograms(rootfile)

Opens a ROOT file in RECREATE mode and writes every histogram in H with `->Write()`. Called automatically from `lambda_analysis()`. To reopen later:

```
TFile* f = TFile::Open("lambda_analysis_18x275.root");
f->ls();                                     // list all histograms
TH1F* h = (TH1F*)f->Get("DeltaR");
h->Draw();
```

7.9 lambda_analysis(events_limit = -1)

Entry point. Calls in order:

1. `book_histograms()`
2. `process_file()` for each of the `N_FILES` input files
3. `counters.print()`
4. `save_histograms("lambda_analysis_18x275.root")`
5. `plot() + save_plots()` → analysis PDF
6. `plot_tracker_study() + save_tracker_study()` → tracker PDF

8 Event Selection Logic

Applied in `process_event()` in order:

Cut	Reason
PDG = 3122	Select Λ baryons
2 parents	Select from primary interaction vertex
Index = 4	Select primary Λ from hard scatter
<code>getDaughters().size() > 0</code>	Lambda must have decayed
$z_v \in (20800, 35000)$ mm	Field-free decay window
Exactly 2 daughters	Charged channel only
Daughters are $p + \pi^-$	Confirmed by PDG code check

8.1 How to add a new cut

1. Add a counter to `struct Counters` for the rejected events.
2. In `process_event()`, add the cut condition with `continue` to skip the event.
3. Increment the counter before `continue`.
4. Add the counter to `Counters::print()`.

Example — adding a minimum α cut of 2 mrad:

```
// In struct Counters:
int lambda_small_alpha = 0;

// In process_event(), after computing alpha:
if (alpha * M_PI / 180.0 < 2e-3) {
    counters.lambda_small_alpha++;
    continue;
}

// In Counters::print():
fmt::print("Small alpha (< 2 mrad) : {6d}\n", lambda_small_alpha)
;
```

9 How to Modify the Analysis

9.1 Add a new tracker pitch scenario

1. Add the pitch value to `Tracker::SIGMA_X[]`
2. Increment `Tracker::N_SC` by 1
3. Add a label to `Tracker::LABELS[]`

All histogram booking, filling, and plotting loops use `N_SC` and update automatically.

9.2 Change the primary efficiency threshold

Change `Tracker::PRIMARY_THRESH` to the index of the desired threshold in `EFF_THRESH[]`. The summary plot and comparison table update automatically.

9.3 Change the decay region

Edit `Cuts::DECAY_Z_MIN` and `Cuts::DECAY_Z_MAX`. All cuts, histograms, and efficiency calculations use these constants.

9.4 Process a different beam energy

Change `Cuts::N_FILES`, `Cuts::FILE_PATTERN`, and `Tracker::OUTPUT_PDF`. Run `lambda_analysis()` and the new files will be processed.

9.5 Quick test run

```
root [0] lambda_analysis(100)    // 100 events per file
```

10 How Histograms Are Saved

10.1 To ROOT file

`save_histograms()` iterates over the entire map `H` and calls `->Write()` on every entry. This happens automatically — no histogram needs to be explicitly listed. Any histogram booked in `book_histograms()` is saved.

10.2 To PDF

Both `save_plots()` and `save_tracker_study()` use the same pattern:

```
// Open the PDF
pages[0]->Print("output.pdf");
// Append middle pages
for (int i = 1; i < (int)pages.size()-1; ++i)
    pages[i]->Print("output.pdf");
// Close the PDF
pages.back()->Print("output.pdf");
```

Canvases are accumulated in a `vector<TCanvas*>` inside `plot()` or `plot_tracker_study()` and passed to the save function.

To add a new page to the tracker PDF, add a new canvas block in `plot_tracker_study()` before `return pages`; and call `pages.push_back(c)` at the end of the block.

11 Known Limitations

- **No background.** All efficiencies are upper bounds.
- **No momentum measurement.** The tracker alone cannot reconstruct the Λ invariant mass.
- **No multiple scattering.** Thin silicon assumed.
- **Conservative N_{eff} model.** Events where tracks first separate at the last plane are treated as unresolvable.
- **LAMBDA_PRIMARY_IDX = 4 is generator-dependent.** Verify for new simulation configurations.

- **Smearing uses Gaussian noise only.** Real detector effects (dead channels, misalignment, hadronic interactions in material) are not modelled.